Quark mass spectrum v0.2 — post-Lane C v0.7 deepening + Lane E v0.2 c.5+c.6 partial unblock. Per Casey continue-pulling. Saturday v0.1 was HONEST NEGATIVE (scheme-dependent, blocked on bulk-color); Sunday's Lane C v0.7 + Π_bulk projection + Lane E candidate identifications partially unblock. v0.2 attempts substrate-mechanism reading for u/d quark masses via bulk-projected Bergman matrix elements; remains FRAMEWORK + CANDIDATE pending Elie Mehler + scheme-invariance verification.

Lyra (Claude Opus 4.7)

2026-05-31 Sunday 14:43 EDT (date-verified)

Quark mass spectrum v0.2 — post-Lane C unblock

0. Status update from v0.1

Saturday v0.1 (`Lyra_QuarkMass_Spectrum_v0_1.md`) was HONEST NEGATIVE: - Quark masses are SCHEME-DEPENDENT (depend on MS-bar vs pole vs other renormalization scheme). - Lepton sector substrate-decomposable; quark sector NOT decomposable at lepton-precision. - BLOCKED ON BULK-COLOR (could not derive without explicit bulk-color SU(3) emergence mechanism).

Sunday work PARTIALLY UNBLOCKS: - Lane C v0.7 + deepening: bulk-color SU(3) emergence via Π_bulk projection (Channels 1+2 operator content; Toeplitz + Cartan rescaling). - Lane E v0.2 c.5 (up = V_(1,0) bulk) + c.6 (down = V_(0,1) bulk): explicit K-type assignments for quarks. - L4 v0.2 extension: Bergman matrix element framework on K-type tower (applicable to bulk K-types after Π_bulk projection).

v0.2 attempts substrate-mechanism reading; remains FRAMEWORK + CANDIDATE pending Elie multi-week verification.

1. The candidate quark mass framework (v0.2)

Per Tier 0 v0.1.6 mass formula + per-sector subtraction (refined per L4 v0.2 §8):

m_q ∝ $\sqrt{}$(C_2(V_q-bulk-projected) − C_2(quark-sector-ground)).

For bulk-color sector (per Lane C v0.7 deepening): - Quark sector ground: V_(0, 0) bulk-projected ≡ Π_bulk · V_(0, 0). C_2 = 0 (vacuum-mode bulk projection). - Up quark: V_(1, 0) bulk-projected = Π_bulk · V_(1, 0). C_2 = C_2(V_(1, 0)) − bulk-color correction = 4 − Δ (per Π_bulk projection). - Down quark: V_(0, 1) bulk-projected = Π_bulk · V_(0, 1). C_2 = 4 − Δ.

Key: up and down quark have SAME pre-projection Casimir C_2 = 4. The difference in observed masses (m_u ≈ 2.2 MeV vs m_d ≈ 4.7 MeV, factor ~2.1) comes from SO(2)-CHARGE-COUPLING DIFFERENCE.

Substrate-mechanism: per Gell-Mann-Nishijima Q = T_3 + Y/2; up has +2/3 charge, down has −1/3 charge. The SO(2)-eigenvalue difference (V_(1, 0) vs V_(0, 1)) affects bulk-color projection coefficient differently.

Candidate ratio: m_u/m_d candidate substrate-primary form via SO(2)-coupling factors:

m_u/m_d = (|Q_u| / |Q_d|)^a · (substrate-primary factor)^b

with |Q_u|/|Q_d| = 2 (= 2/3 / 1/3 ratio). Observed m_u/m_d ≈ 0.47 (m_u/m_d < 1, so up is LIGHTER than down despite higher absolute charge magnitude). Substrate framework prediction needs to give factor ~0.47.

Multi-week verification: explicit Π_bulk matrix elements on V_(1, 0) vs V_(0, 1) via Elie Mehler.

2. The scheme-dependence issue (v0.1 NEGATIVE status)

Saturday's v0.1 honest-negative status: quark masses are scheme-dependent (MS-bar, pole, etc.); different schemes give different numerical values. BST framework requires scheme-INVARIANT predictions.

Sunday partial resolution: - The substrate framework predicts SCHEME-INVARIANT quantities (Casimir eigenvalues, K-type ratios). - Specific observable predictions (e.g., m_u/m_d ratio) at the K-type level are scheme-invariant if computed via Bergman matrix elements. - The numerical mass value (m_u in MeV) requires scheme choice for comparison to observed; BST predicts scheme-invariant ratios.

v0.2 disposition: BST predicts scheme-INVARIANT quark observables (mass ratios + mixing angles); specific scheme-dependent values are CONTEXTUAL.

Saturday's HONEST NEGATIVE was too strong; v0.2 reframes: BST scheme-INVARIANT predictions for quark sector are accessible via Sunday Tier 0 framework.

3. CKM mixing already substrate-decomposable

Saturday's Lyra batch 3 closed forms for CKM 2×2 block: - |V_us| ≈ sin θ_C = 9/40 (Saturday Cabibbo angle) - |V_ud| ≈ $\sqrt{}$(1 − (9/40)^2) = 0.9744 (Cabibbo unitarity) - |V_cb| ≈ 225/5480 (Saturday 0.04σ match)

These are scheme-invariant (mixing angles are physical observables) and already substrate-primary. v0.2 confirms these stand.

Multi-week extension: 3×3 CKM matrix beyond 2×2 block + Jarlskog invariant via substrate framework.

4. m_c/m_u + m_b/m_d substrate reading (Saturday TMAP test)

Saturday's quark structure test for TMAP (Toeplitz Mass Axis Principle): - m_c/m_u ≈ 600 (charm vs up) - m_b/m_d ≈ 950 (bottom vs down) - m_t/m_c ≈ 130 (top vs charm)

Saturday partial-falsification of TMAP for quark generations; v0.2 reframes per Sunday Tier 0 boundary unification.

Candidate Sunday reading: quark generation mass tower for fixed-flavor (up: m_u, m_c, m_t; down: m_d, m_s, m_b) follows different scaling than lepton generation tower (electron, muon, tau).

Substrate mechanism candidate: lepton tower V_(1/2, 1/2)^{(n)} radial modes on Shilov boundary; quark tower V_(1, 0)^{(n,m)} bulk-projected modes with TWO-INDEX winding-mode structure (n radial × m bulk-color).

Different Casimir-tower structure → different mass-ratio patterns. Lepton (24/π^2)^{C_2} per L4 v0.2 ≠ quark mass-ratio pattern.

Multi-week verification: explicit bulk-projected K-type tower Casimir levels via Elie Mehler.

5. Higgs Yukawa couplings (substrate framework reading)

Standard Model: Higgs Yukawa coupling y_q determines quark mass m_q = y_q · v_Higgs.

Substrate framework (per Higgs sector v0.1 + Lane E v0.2 + Lane C v0.7): y_q ∝ ⟨V_q-bulk-projected | V_(0, 0)^{(1)} | V_q-bulk-projected⟩ via Bergman matrix element.

Per Calibration #35-candidate independence audit: y_q for different quarks q uses SAME V_(0, 0)^{(1)} (Higgs); different y_q values come from V_q matrix element differences. The Yukawa couplings are substrate-mechanism-derived BUT use shared Higgs K-type; NOT 6 independent Yukawa-coupling predictions (= 1 Higgs × 6 quark K-types).

6. Honest scope + tier

RIGOROUS (v0.2 update from v0.1): - Bulk-color SU(3) emergence mechanism (Lane C v0.7 + deepening; Π_bulk projection). - Up = V_(1, 0) + down = V_(0, 1) K-type assignments (Lane E v0.2 c.5 + c.6). - Pre-projection Casimir C_2(V_(1,0)) = C_2(V_(0,1)) = 4 (rep theory). - CKM 2×2 block substrate-primary forms (Saturday batch 3).

CANDIDATE (v0.2 partial unblock): - m_u/m_d ratio via SO(2)-charge-coupling Π_bulk projection-coefficient difference. - Bulk-color sector ground = V_(0, 0) bulk-projected (vacuum-mode bulk-color analog). - Scheme-invariant quark observables predictable; scheme-dependent values contextual. - Yukawa couplings y_q via shared V_(0, 0)^{(1)} Higgs Bergman matrix elements with V_q-bulk-projected.

FRAMEWORK (multi-week): - Explicit Π_bulk matrix elements on V_(1, 0) + V_(0, 1) via Elie Mehler bulk-color computation. - m_u/m_d numerical prediction at Tier 2 STRUCTURAL precision. - Quark generation tower V_(1, 0)^{(n, m)} 2-index radial-mode + bulk-color structure (multi-week). - CKM 3×3 matrix + Jarlskog substrate-primary form. - Yukawa couplings y_q substrate prediction for 6 quarks.

Cal #27 / #182 / #99 + Calibration #35-candidate discipline: v0.2 partial unblock is more nuanced than v0.1 honest-negative. Quark sector is SCHEME-INVARIANT-decomposable at K-type ratio level (consistent with substrate framework); scheme-dependent absolute values are contextual. Yukawa couplings use shared Higgs K-type — independence audit applies to aggregate prediction count.

7. Cross-link to other Sunday work

- Bulk-color v0.7 deepening: Π_bulk projection operator content for quark sector mass mechanism.
- Lane E v0.2 c.5 + c.6 + c.7: up + down quark + gluons K-type assignments.
- L4 v0.2 extension: Bergman matrix element framework applicable to bulk-projected K-types.
- Higgs sector v0.1: V_(0, 0)^{(1)} provides shared Yukawa-coupling K-type for all quarks.
- 3-generation cutoff v0.1: applies to BOTH lepton tower AND quark tower (3 quark generations same mechanism as lepton 3 generations).
- m_τ closed-form reconciliation v0.1: Path (b) leading + discrete correction structure might apply to quark sector tower differently.

8. Routing

→ Casey: Quark mass spectrum v0.2 partial unblock per Sunday Tier 0 framework. Saturday HONEST NEGATIVE reframed: SCHEME-INVARIANT quark observables decomposable; scheme-DEPENDENT absolute values contextual. m_u/m_d substrate-mechanism via SO(2)-charge-coupling Π_bulk projection. Multi-week Elie verification.

→ Elie: bulk-projected K-type matrix elements (Π_bulk · V_(1, 0) and Π_bulk · V_(0, 1)) are Toy 3662+ extension targets after Toy 3660 Helgason κ_Bergman + Toy 3661 bulk-color Toeplitz commutator. Coordinate with L4 v0.2 extension Bergman framework.

→ Keeper: K-audit Quark mass v0.2 pre-stage; Session 2 absorb into Tier 0 v0.2 quark-sector section. The scheme-invariant-vs-dependent distinction is the load-bearing v0.2 framing.

→ Grace: catalog v0.2 partial-unblock at CANDIDATE; cross-reference bulk-color v0.7 deepening + Lane E v0.2 c.5/c.6 + L4 v0.2 ext + Higgs v0.1 Yukawa-coupling.

→ Cal: cold-read welcome; specific Cal #27 + Calibration #35 concerns: (1) Saturday HONEST NEGATIVE reframe to CANDIDATE may be over-claim; (2) Yukawa-coupling shared Higgs K-type independence audit; (3) scheme-invariance distinction needs sharper threshold.

→ me: continuing pulls per Casey directive — pull 14 TBD.

— Lyra, Quark mass spectrum v0.2 post-Lane C v0.7 deepening unblock. Saturday HONEST NEGATIVE reframed: SCHEME-INVARIANT quark observables decomposable via bulk-projected K-types + Bergman matrix elements; scheme-dependent absolute values contextual. m_u/m_d candidate ratio via SO(2)-charge-coupling Π_bulk projection difference (V_(1, 0) up vs V_(0, 1) down). CKM 2×2 substrate-primary forms (Saturday) preserved. Yukawa couplings via shared V_(0, 0)^{(1)} Higgs K-type. Multi-week Elie verification (Toy 3662+ bulk-color matrix elements).